

Endocrine Glands: Secretion and Action of Hormones

Human Physiology

Biology Teaching Resources

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Outline

11.1 Endocrine Glands and Hormones

11.2 Mechanisms of Hormone Action

11.3 Pituitary Gland

11.4 Adrenal Glands

11.5 Thyroid and Parathyroid Glands

11.6 Pancreas and Other Endocrine Glands

11.7 Autocrine and Paracrine Regulation

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11.1 Endocrine Glands and Hormones

- **Endocrine glands** (内分泌腺) are ductless glands that secrete **hormones** (激素) into blood
- **Hormones** travel through bloodstream to reach **target cells** (靶细胞) with specific **receptors** (受体)
- Unlike **exocrine glands** (外分泌腺) which have ducts
- Examples: pancreas, thyroid, adrenal, pituitary, gonads



11.1 Endocrine Glands and Hormones

- Major endocrine glands and their locations throughout the body
- Some organs have both endocrine and exocrine functions (e.g., pancreas)
- Many organs have secondary endocrine functions (heart, kidneys, liver, adipose tissue)

11.1 Endocrine Glands and Hormones

- Comprehensive overview of endocrine glands, their locations, hormones secreted, target organs, and primary effects
- Demonstrates the widespread influence of the endocrine system on body functions



11.1 Endocrine Glands and Hormones

- Hormones are classified into four main chemical categories based on structure
- **1. Amines** (胺类): From tyrosine or tryptophan (epinephrine, thyroid hormones, melatonin)
- **2. Polypeptides/Proteins** (多肽/蛋白质): Chains of amino acids (ADH, insulin, growth hormone)
- **3. Glycoproteins** (糖蛋白): Proteins with carbohydrates (FSH, LH)
- **4. Steroids** (类固醇): Derived from cholesterol (testosterone, estradiol, cortisol)

11.1 Endocrine Glands and Hormones

- Examples of polypeptide and glycoprotein hormones from various glands
- Shows diversity of hormone structures and functions



11.1 Endocrine Glands and Hormones

- **Steroid hormones** all derived from cholesterol through enzymatic pathways
- **Progesterone** (孕酮) is the common precursor for all other steroid hormones
- Different zones of adrenal cortex and gonads produce different steroids
- Androgens can be converted to estrogens



11.1 Endocrine Glands and Hormones

- **Thyroid hormones** are unique: amine-derived but lipid-soluble like steroids
- T_4 (thyroxine) has 4 iodine atoms; T_3 (triiodothyronine) has 3
- Secreted in 9:1 ratio, but T_3 is 3-4× more potent
- Both derived from tyrosine amino acid with iodine additions



11.1 Endocrine Glands and Hormones

- **Prehormones** (前激素) are inactive precursors requiring modification
- Examples: proinsulin → insulin; vitamin D₃ → active 1,25-dihydroxyvitamin D₃
- Allows regulation of hormone activity at multiple levels



11.1 Endocrine Glands and Hormones

- Hormones can interact in three ways:
- **1. Synergistic** (协同): Work together (additive or complementary effects)
- **2. Permissive** (允许): One enhances response to another
- **3. Antagonistic** (拮抗): Opposite effects (insulin vs. glucagon)

11.1 Endocrine Glands and Hormones

- **Physiological concentrations** (生理浓度): Normal ranges produce appropriate responses
- **Pharmacological concentrations** (药理浓度): High doses may cause abnormal effects
- **Priming** (启动): Small amounts increase receptor sensitivity (**upregulation**, 上调)
- **Desensitization** (脱敏): Prolonged high exposure decreases response (**downregulation**, 下调)

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11.2 Mechanisms of Hormone Action

- Hormone mechanisms depend on chemical solubility:
- **Lipophilic** (脂溶性): Steroids and thyroid hormones cross membranes, bind **nuclear receptors** (核受体)
- **Hydrophilic** (水溶性): Peptides and catecholamines cannot cross membranes, bind **surface receptors**, use **second messengers** (第二信使)

11.2 Mechanisms of Hormone Action

- **Steroid hormone mechanism:** Cross membrane → bind nuclear receptor → receptor-hormone complex binds DNA → regulate gene transcription
- Effects are **genomic** (基因组的):
Change gene expression, take hours to days
- Examples: cortisol, estrogen, testosterone



11.2 Mechanisms of Hormone Action

- **Nuclear receptors** have two functional domains:
- **Ligand-binding domain** (配体结合域): Binds the hormone
- **DNA-binding domain** (DNA 结合域): Binds to **hormone-response elements** (HRE) on DNA
- Receptors form **dimers** (二聚体) to activate transcription



11.2 Mechanisms of Hormone Action

- **Thyroid hormone mechanism:** T_4 enters cell $\rightarrow$ converted to T_3 $\rightarrow$ T_3 binds nuclear receptor $\rightarrow$ activates transcription
- T_3 is the more active form
- Increases cellular metabolism and oxygen consumption



11.2 Mechanisms of Hormone Action

- Thyroid hormone receptor forms **heterodimer** (异源二聚体) with retinoic acid receptor
- Both must bind to DNA for full activation
- Demonstrates complexity of gene regulation



11.2 Mechanisms of Hormone Action

- **cAMP second messenger system:**
Most common pathway
- Hormone binds receptor → activates **G-protein** (G 蛋白) → activates **adenylate cyclase** → produces **cAMP** → activates **protein kinase A** (PKA) → phosphorylates proteins
- **Amplification** (放大): One hormone → many cAMP → many phosphorylated proteins



11.2 Mechanisms of Hormone Action

- Step-by-step sequence of cAMP second messenger system
- Shows temporal progression and key enzymes involved
- **Phosphodiesterase** (磷酸二酯酶) terminates signal by degrading cAMP



11.2 Mechanisms of Hormone Action

- **Calcium second messenger system:** Hormone $\rightarrow$ G-protein $\rightarrow$ **phospholipase C** $\rightarrow$ cleaves $\text{PIP}_2 \rightarrow \text{IP}_3 + \text{DAG}$
- **IP_3** releases Ca^{2+} from ER
- $\text{Ca}^{2+} + \text{DAG}$ activate **protein kinase C** (PKC)
- Ca^{2+} also binds **calmodulin** (钙调蛋白) to activate other enzymes



11.2 Mechanisms of Hormone Action

- Detailed sequence of calcium second messenger pathway
- IP_3 and DAG serve as separate branches from same precursor (PIP_2)
- Both pathways converge to produce cellular responses



11.2 Mechanisms of Hormone Action

- Some hormones use **multiple second messengers** simultaneously
- **Epinephrine** on liver: β -receptors $\rightarrow$ cAMP; α -receptors $\rightarrow$ Ca^{2+}
- Both pathways promote glycogen breakdown
- Demonstrates redundancy and integration in signaling



11.2 Mechanisms of Hormone Action

- **Insulin receptor: Tyrosine kinase**
(酪氨酸激酶) type
- Insulin binding → receptor dimerization → **autophosphorylation**
(自磷酸化) → phosphorylates other proteins
- Initiates complex signaling cascades including MAP kinase pathway
- Produces both rapid and long-term effects



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11.3 Pituitary Gland

- **Pituitary gland (hypophysis, 垂体):**
“Master gland” controlling other endocrine glands
- Two lobes with different structures and functions:
- **Anterior pituitary (adenohypophysis, 前叶):** Glandular tissue
- **Posterior pituitary (neurohypophysis, 后叶):** Nervous tissue



11.3 Pituitary Gland

- Anterior pituitary hormones and their targets
- **Tropic hormones** (促激素): TSH, ACTH, FSH, LH regulate other glands
- **Direct hormones**: GH and prolactin act on target tissues



11.3 Pituitary Gland

- Overview of anterior pituitary control over other endocrine organs
- Demonstrates hierarchical organization of endocrine system
- Not all endocrine glands are controlled by pituitary



11.3 Pituitary Gland

- **Posterior pituitary** stores and releases hormones made in hypothalamus
- **ADH** (**vasopressin**, 抗利尿激素): Water reabsorption in kidneys
- **Oxytocin** (催产素): Uterine contractions, milk ejection
- Neurons from hypothalamus extend into posterior pituitary



11.3 Pituitary Gland

- **Hypothalamo-hypophyseal portal system** (下丘脑-垂体门脉系统)
- Specialized blood vessels directly connect hypothalamus to anterior pituitary
- Allows efficient delivery of **releasing hormones** and **inhibiting hormones**
- Hypothalamus controls anterior pituitary secretion



11.3 Pituitary Gland

- Hypothalamic hormones regulating anterior pituitary
- **Releasing hormones**: TRH, CRH, GnRH, GHRH stimulate secretion
- **Inhibiting hormones**: Somatostatin, PIH (dopamine) suppress secretion



11.3 Pituitary Gland

- **Hypothalamus-Pituitary-Thyroid Axis**: Classic negative feedback loop
- Hypothalamus → TRH → Pituitary → TSH → Thyroid → T_3 , T_4
- T_3 , T_4 inhibit TRH and TSH (**negative feedback**, 负反馈)
- Maintains stable thyroid hormone levels



11.3 Pituitary Gland

- **Hypothalamus-Pituitary-Gonad Axis**
- Hypothalamus → GnRH → Pituitary → FSH, LH → Gonads → Sex steroids
- Sex steroids usually inhibit GnRH, FSH, LH
- Exception: Positive feedback during ovulation (estrogen surge → LH surge)



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11.4 Adrenal Glands

- **Adrenal glands** (肾上腺): Paired glands atop kidneys
- Two regions: **Cortex** (皮质, outer, 80%) and **Medulla** (髓质, inner, 20%)
- Cortex has 3 zones: **zona glomerulosa**, **fasciculata**, **reticularis**
- Different zones secrete different steroid hormones



11.4 Adrenal Glands

- Biosynthetic pathways for adrenal cortex steroids
- All derived from cholesterol
- **Zona glomerulosa:**
Mineralocorticoids (aldosterone)
- **Zona fasciculata/reticularis:**
Glucocorticoids (cortisol), sex steroids



11.4 Adrenal Glands

- **Mineralocorticoids** (盐皮质激素): Primarily **aldosterone** (醛固酮)
- Functions: Increases Na^+ reabsorption, K^+ secretion in kidneys
- Increases blood volume and blood pressure
- Regulated by angiotensin II and K^+ levels
- Excess: **Conn's syndrome** - hypertension
- Deficiency: **Addison's disease** - hypotension

11.4 Adrenal Glands

- **Glucocorticoids** (糖皮质激素): Primarily **cortisol** (皮质醇)
- Functions: Stimulates **gluconeogenesis** (糖异生), raises blood glucose
- Promotes protein breakdown, lipolysis
- **Anti-inflammatory** (抗炎) and **immunosuppressive** (免疫抑制)
- Excess: **Cushing's syndrome** - moon face, obesity, hyperglycemia
- Deficiency: **Addison's disease** - hypoglycemia, weakness, hyperpigmentation

11.4 Adrenal Glands

- **Adrenal medulla** secretes **catecholamines** (儿茶酚胺)
- 80% epinephrine, 20% norepinephrine
- **Fight-or-flight response**: Increases heart rate, blood glucose, redirects blood flow
- Binds **α -receptors** (excitatory, vasoconstriction) and **β -receptors** (usually inhibitory except heart)

11.4 Adrenal Glands

- **Stress** (应激) activates **HPA axis** (hypothalamus-pituitary-adrenal axis)
- Stress → CRH → ACTH → Cortisol
- Cortisol provides energy and reduces inflammation
- **Negative feedback** prevents excessive response
- Chronic stress → cortisol excess, immunosuppression



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11.5 Thyroid and Parathyroid Glands

- **Thyroid gland** (甲状腺): Butterfly-shaped in neck, anterior to trachea
- Two lobes connected by **isthmus**
- Secretes thyroid hormones and calcitonin



11.5 Thyroid and Parathyroid Glands

- **Thyroid follicles** (甲状腺滤泡):
Spherical structures
- **Follicular cells** surround **colloid** (胶质) containing **thyroglobulin** (甲状腺球蛋白)
- **C cells** (**parafollicular cells**) secrete **calcitonin** (降钙素)



11.5 Thyroid and Parathyroid Glands

- Thyroid hormone synthesis is complex:
- (1) Iodide actively transported (Na^+/I^- symporter)
- (2) Oxidized to iodine by thyroid peroxidase
- (3) Attached to tyrosine in thyroglobulin (MIT, DIT)
- (4) Coupling: $\text{MIT} + \text{DIT} \rightarrow \text{T}_3$ $\text{DIT} + \text{DIT} \rightarrow \text{T}_4$
- (5) TSH triggers endocytosis and hormone release



11.5 Thyroid and Parathyroid Glands

- Thyroid hormone functions:
- Increase **basal metabolic rate** (BMR, 基础代谢率): oxygen consumption, heat production
- Essential for normal brain development (fetus/infant)
- Promote growth, skeletal maturation
- Increase protein synthesis, lipid metabolism
- Permissive for catecholamines

11.5 Thyroid and Parathyroid Glands

- **Endemic goiter** (地方性甲状腺肿)
from iodine deficiency
- Thyroid enlargement visible in neck
- Most common cause of
hypothyroidism worldwide



11.5 Thyroid and Parathyroid Glands

- Mechanism of goiter formation:
- Low dietary iodine → low thyroid hormones → reduced negative feedback
- → elevated TSH → thyroid hypertrophy → goiter
- Can be prevented by iodized salt



11.5 Thyroid and Parathyroid Glands

- **Hypothyroidism** (甲状腺功能减退): Decreased BMR, cold intolerance, weight gain, fatigue
- **Myxedema** (黏液性水肿): Edema in adults
- **Cretinism** (呆小症): Mental retardation in infants
- **Hyperthyroidism** (甲状腺功能亢进): Increased BMR, heat intolerance, weight loss, nervousness
- **Exophthalmos** (突眼): Eye protrusion in Graves' disease



11.5 Thyroid and Parathyroid Glands

- Comparison of hypothyroidism vs. hyperthyroidism
- Opposite effects on metabolism, temperature regulation, weight, heart rate



11.5 Thyroid and Parathyroid Glands

- **Parathyroid glands** (甲状旁腺):
Usually 4 small glands embedded in posterior thyroid
- Secrete **PTH** (parathyroid hormone, 甲状旁腺激素)



11.5 Thyroid and Parathyroid Glands

- **PTH** raises blood **calcium** (Ca^{2+}) through three mechanisms:
- **1. Bone**: Stimulates osteoclasts → bone resorption → Ca^{2+} release
- **2. Kidneys**: Increases Ca^{2+} reabsorption, decreases PO_4^{3-} reabsorption, activates vitamin D_3
- **3. Intestines** (via vitamin D_3): Increases Ca^{2+} absorption



11.5 Thyroid and Parathyroid Glands

- PTH regulation by **negative feedback**:
- Low Ca^{2+} → stimulates PTH
- High Ca^{2+} → inhibits PTH
- Disorders: Hyperparathyroidism (kidney stones, bone loss), Hypoparathyroidism (muscle spasms, tetany)
- **Calcitonin** from C cells lowers Ca^{2+} but minor role in humans

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11.6 Pancreas and Other Endocrine Glands

- Pancreas: Both exocrine (digestive enzymes) and endocrine functions
- **Pancreatic islets** (胰岛, islets of Langerhans): 1-2% of pancreas
- **α cells** secrete **glucagon**; **β cells** secrete **insulin**



11.6 Pancreas and Other Endocrine Glands

- **Insulin** mechanism: Binding activates receptor → intracellular signaling
- → GLUT4 transporters move to membrane → increased glucose uptake
- Primarily affects muscle and adipose tissue



11.6 Pancreas and Other Endocrine Glands

- **Insulin** (胰岛素): Anabolic hormone, secreted after meals
- Functions: Promotes glucose uptake, **glycogenesis** (糖原合成), **lipogenesis** (脂肪合成), protein synthesis
- Inhibits gluconeogenesis, glycogenolysis, lipolysis
- Net effect: Lowers blood glucose, promotes energy storage

11.6 Pancreas and Other Endocrine Glands

- **Glucagon** (胰高血糖素): Catabolic hormone, secreted during fasting
- Functions: Stimulates **glycogenolysis** (糖原分解), **gluconeogenesis**, **lipolysis**
- Net effect: Raises blood glucose, mobilizes energy

11.6 Pancreas and Other Endocrine Glands

- **Glucose homeostasis:** Maintained by antagonistic effects of insulin and glucagon
- After meal: High glucose $\rightarrow$ insulin $\uparrow$, glucagon $\downarrow$ $\rightarrow$ uptake and storage
- During fasting: Low glucose $\rightarrow$ glucagon $\uparrow$, insulin $\downarrow$ $\rightarrow$ glucose release
- Normal range: 70-110 mg/dL (fasting)



11.6 Pancreas and Other Endocrine Glands

- **Diabetes mellitus** (糖尿病): Chronic hyperglycemia
- **Type 1**: Autoimmune β cell destruction, absolute insulin deficiency, childhood onset, requires insulin
- **Type 2**: Insulin resistance + relative deficiency, adult onset, obesity-related
- Complications: Cardiovascular disease, nephropathy, retinopathy, neuropathy

11.6 Pancreas and Other Endocrine Glands

- **Pineal gland** (松果体) secretes **melatonin** (褪黑素)
- Synthesized from tryptophan → serotonin → melatonin



11.6 Pancreas and Other Endocrine Glands

- Melatonin secretion controlled by **suprachiasmatic nucleus** (SCN, 视交叉上核)
- Light inhibits; darkness stimulates
- **Circadian rhythm** (昼夜节律): High at night, low during day
- Functions: Regulates sleep-wake cycles, seasonal reproduction



11.6 Pancreas and Other Endocrine Glands

- **GI tract hormones**: Gastrin (胃泌素), secretin (促胰液素), CCK (胆囊收缩素)
- Regulate digestion and appetite
- **Gonads** (性腺): Testes secrete testosterone; ovaries secrete estrogens and progesterone
- Regulate reproduction and secondary sex characteristics
- **Placenta** (胎盘): Secretes hCG, estrogens, progesterone during pregnancy

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11.7 Autocrine and Paracrine Regulation

- Three types of chemical signaling:
- **Endocrine** (内分泌): Hormones in blood to distant targets
- **Autocrine** (自分泌): Act on same cells that secrete them
- **Paracrine** (旁分泌): Act on nearby cells



11.7 Autocrine and Paracrine Regulation

- **Growth factors** (生长因子): EGF (skin), NGF (neurons), VEGF (**angiogenesis**, 血管生成)
- **Cytokines** (细胞因子): Interleukins (白细胞介素), interferons (干扰素), TNF (肿瘤坏死因子)
- **Nitric oxide** (NO, 一氧化氮): Vasodilation, short half-life, blood pressure regulation

11.7 Autocrine and Paracrine Regulation

- **Prostaglandins** (前列腺素) and **eicosanoids** (二十烷酸衍生物)
- 20-carbon fatty acids from **arachidonic acid** (花生四烯酸)
- Two pathways: **COX** → prostaglandins/thromboxanes;
Lipoxygenase → leukotrienes



11.7 Autocrine and Paracrine Regulation

- Prostaglandin functions:
- **Inflammation** (炎症): Increase blood flow, permeability, pain
- **Fever** (发热): PGE_2 raises body temperature
- **Blood clotting**: Thromboxane A_2 promotes; PGI_2 inhibits
- Smooth muscle contraction: Uterus (labor), bronchi
- Gastric protection: Inhibit acid, increase mucus

11.7 Autocrine and Paracrine Regulation

- **NSAIDs** (非甾体抗炎药): Aspirin, ibuprofen, naproxen
- Inhibit **COX** enzymes → reduce prostaglandins
- Effects: Reduce inflammation, pain, fever
- Side effects: Gastric ulcers, bleeding
- **COX-2 inhibitors**: Fewer GI effects but cardiovascular risk
- Leukotrienes: Allergic responses, asthma (bronchoconstriction)